

Quantum Hardware

Yousung Kang,
yskang@megazone.com
Quantum Innovation Lab

INDEX

1. Quantum Computers

1. Superconducting Qubits

1. Quantum Annealers

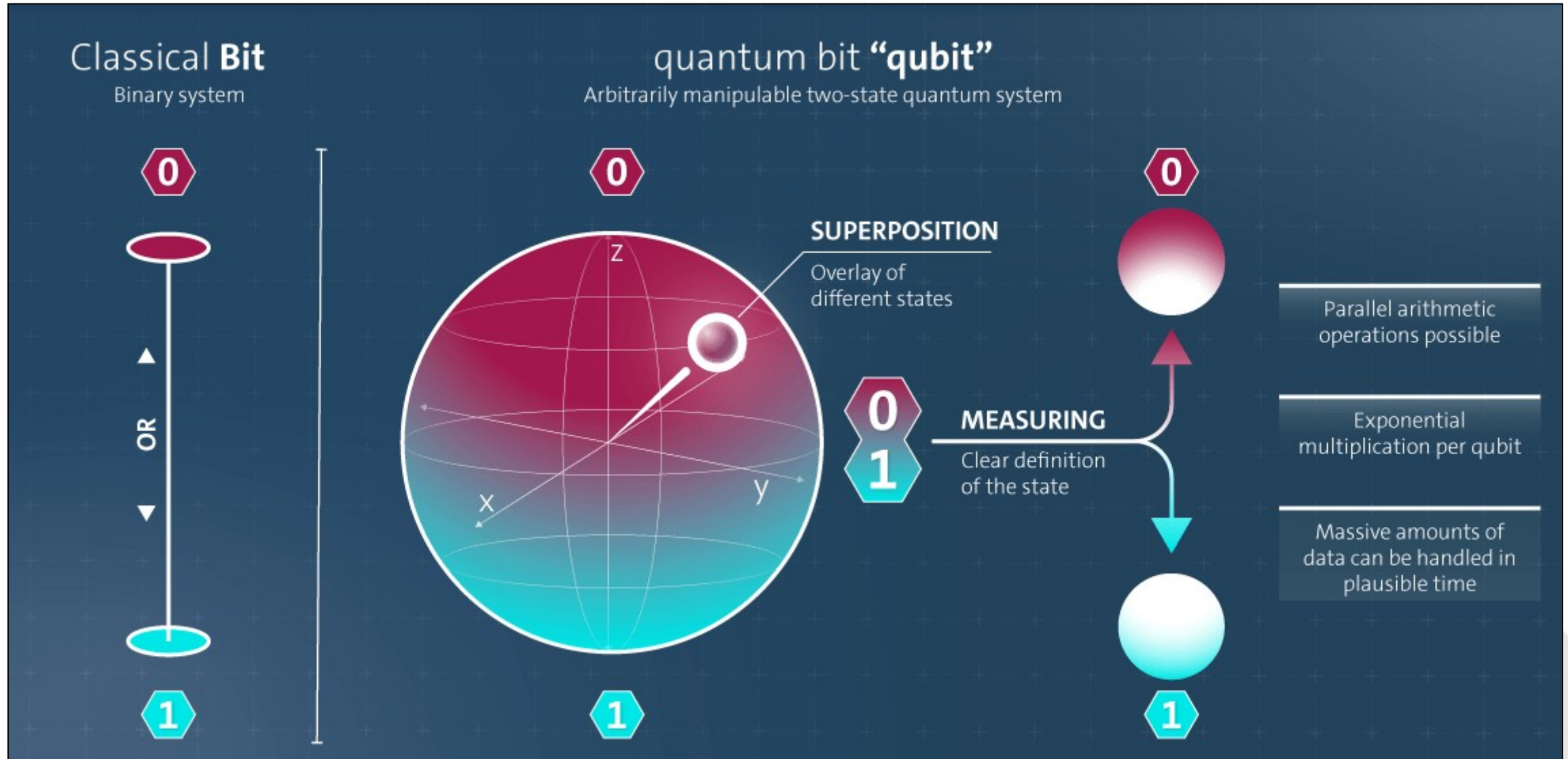
1. Ion Trap Qubits

1. Neutral Atom Qubits

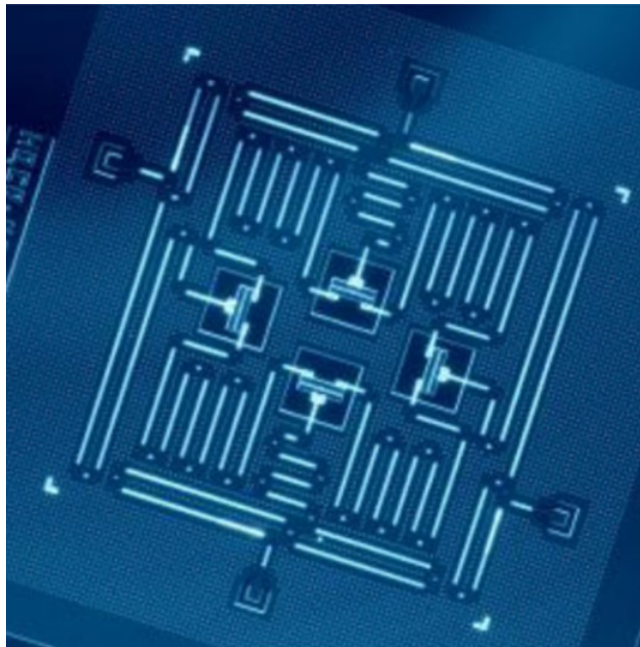
1. Quantum Computers

Q u a n t u m
Innovation Laboratory

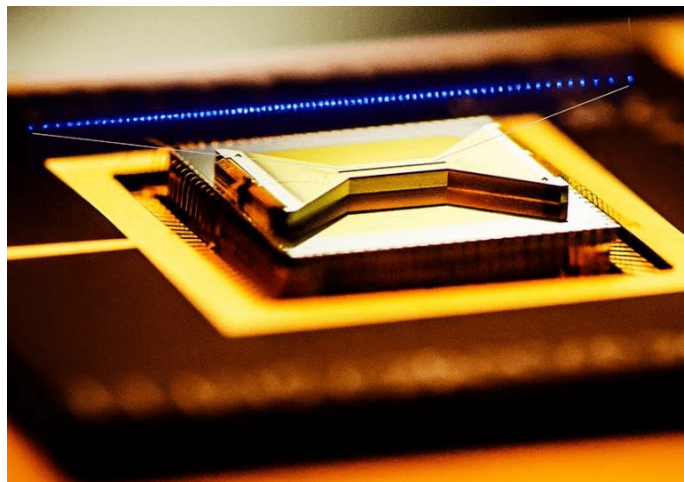
Bit vs Qubit



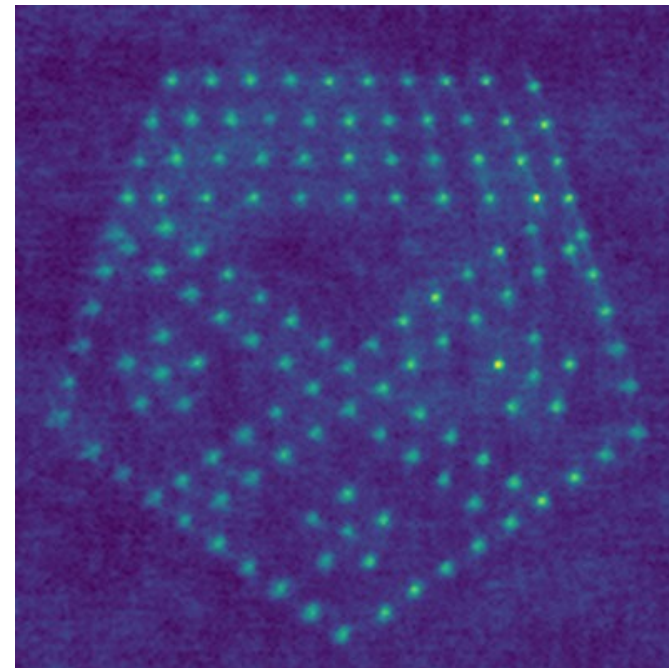
Qubits



1973, 2012, 2025



1989, 2012

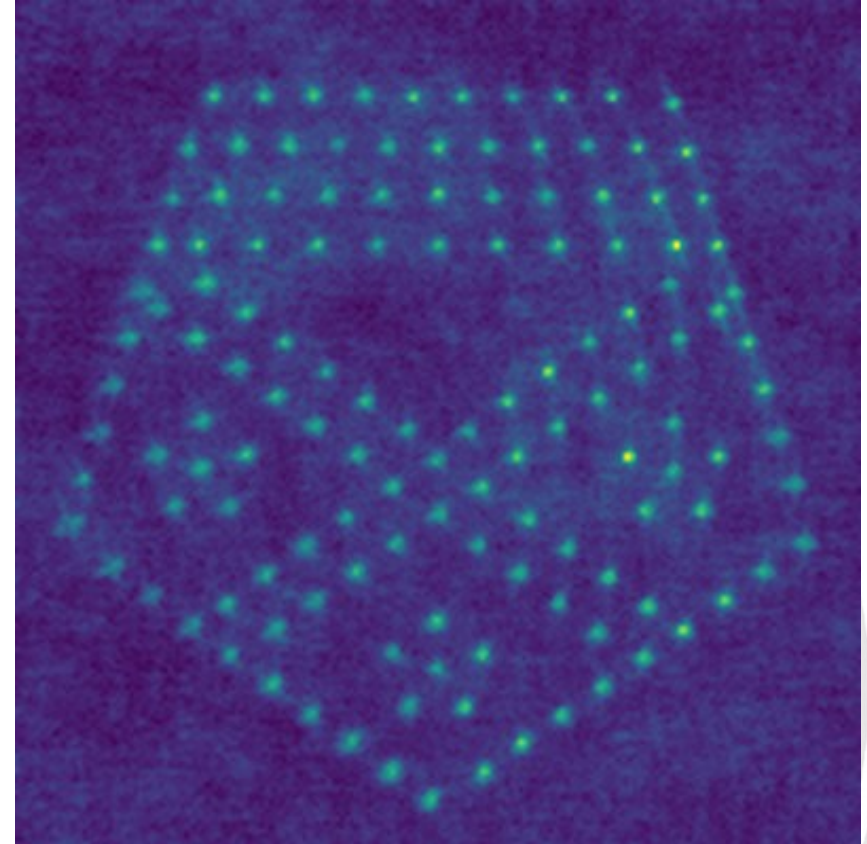


2018

Analog Computing: Specialist

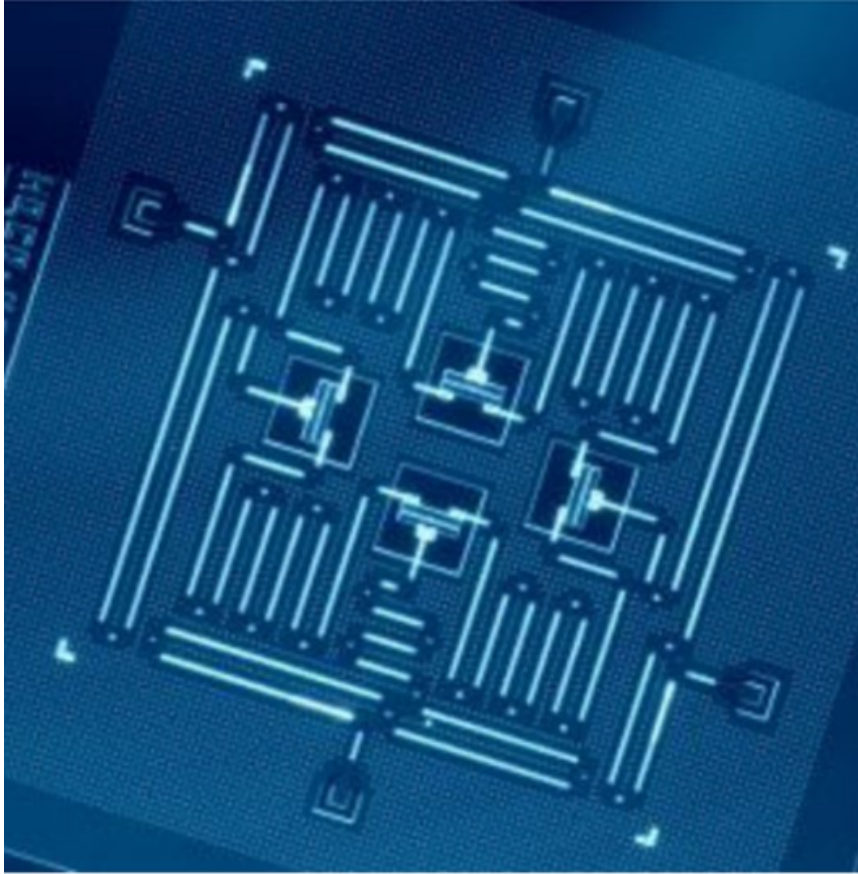


Quantum Annealer
QUBO Specialist

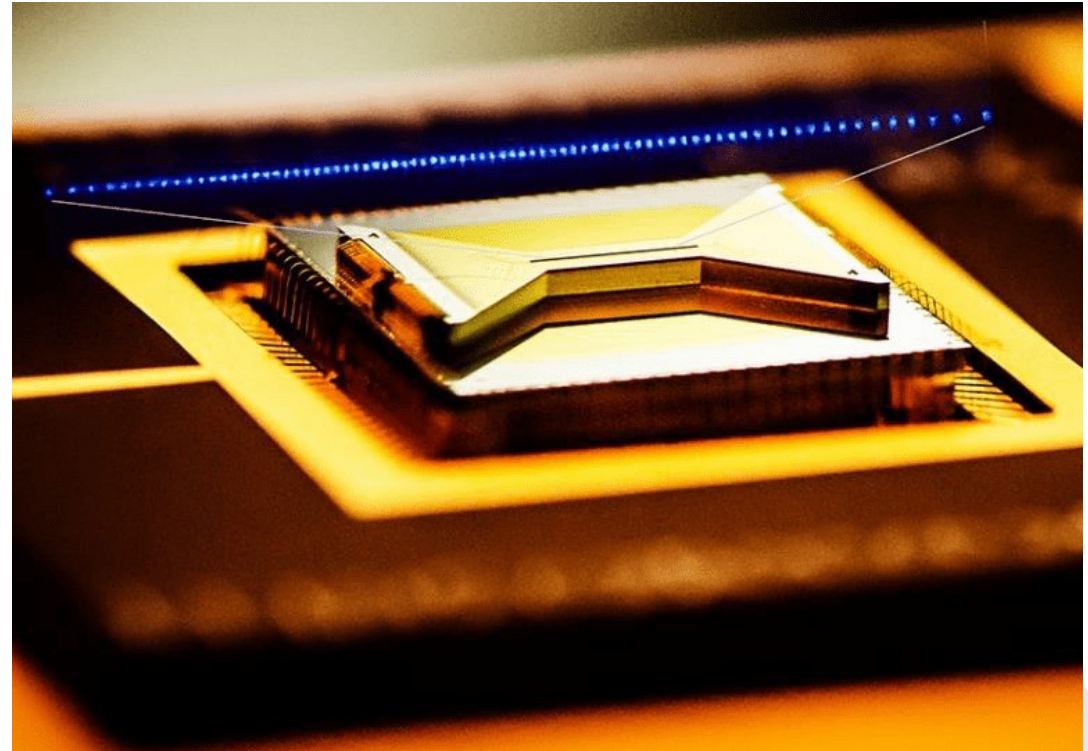


Neutral Atom
MIS Specialist

Gate-base Computing: Generalist



Superconducting



Ion Trap

Liers. T

Teacher:

- $1 - 2 = ?$?: 초등학생
- $\sqrt{-1} = ?$?: 중학생
- $e^{\pi i} = ?$?: 고등학생
- $\int_{-1}^1 \frac{1}{x} dx = ?$?: 고등학생

이것들은 할 수 없어요

Temperature:

- 초등학생: 물질의 차갑고 뜨거운 정도를 숫자로 정의한 것
- 고등학생: 입자들의 평균 운동 에너지 상태를 나타내는 물리량
- 통계역학: 특정 에너지 상태에 대한 입자의 존재확률을 결정하는 핵심 변수

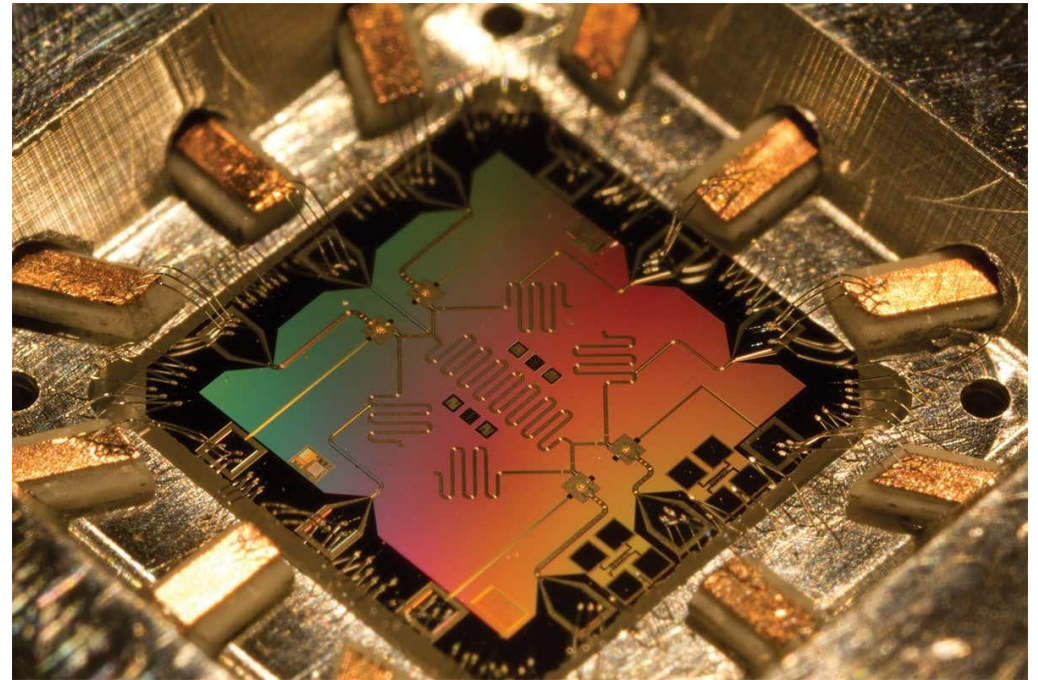
$$P_i \propto e^{-\frac{E_i}{k_B T}}$$

- 열 교환 = 양자컴퓨터의 에러 그 자체

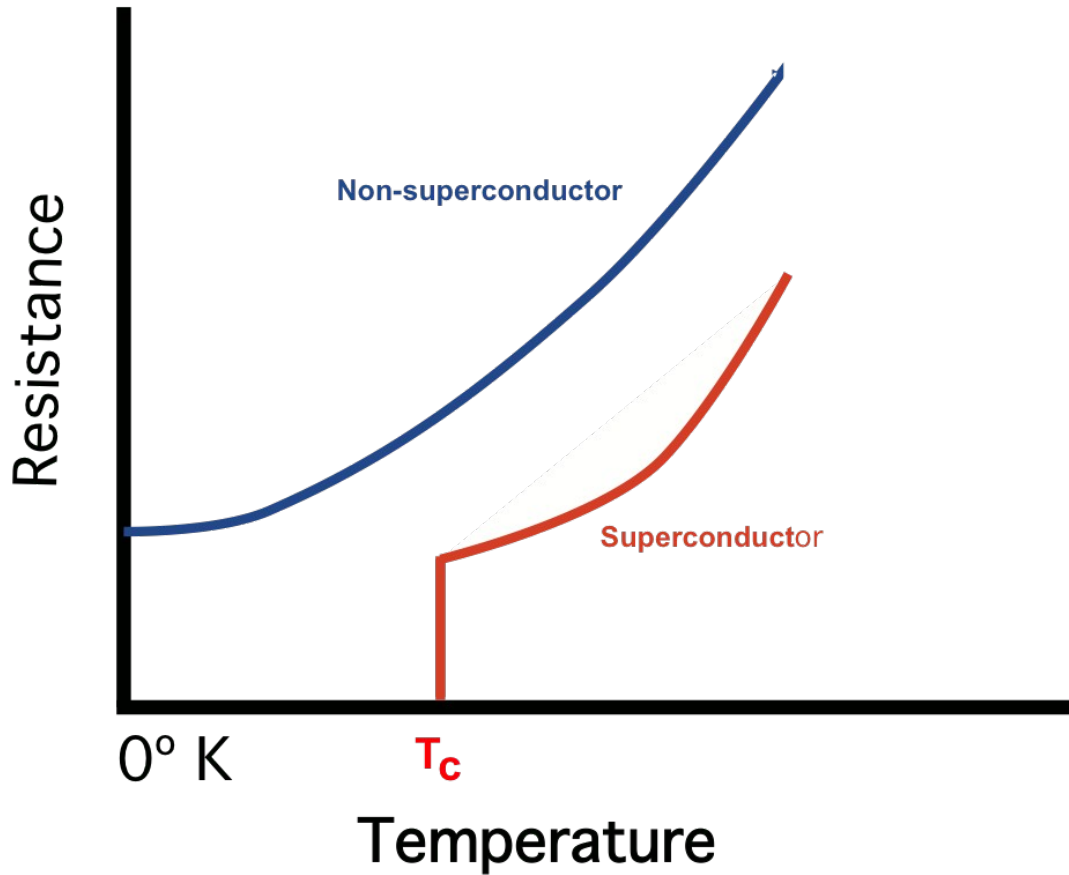
2. Superconducting Qubits

Q u a n t u m
Innovation Laboratory

Superconducting Quantum Computer



Superconductivity depends on Temperature

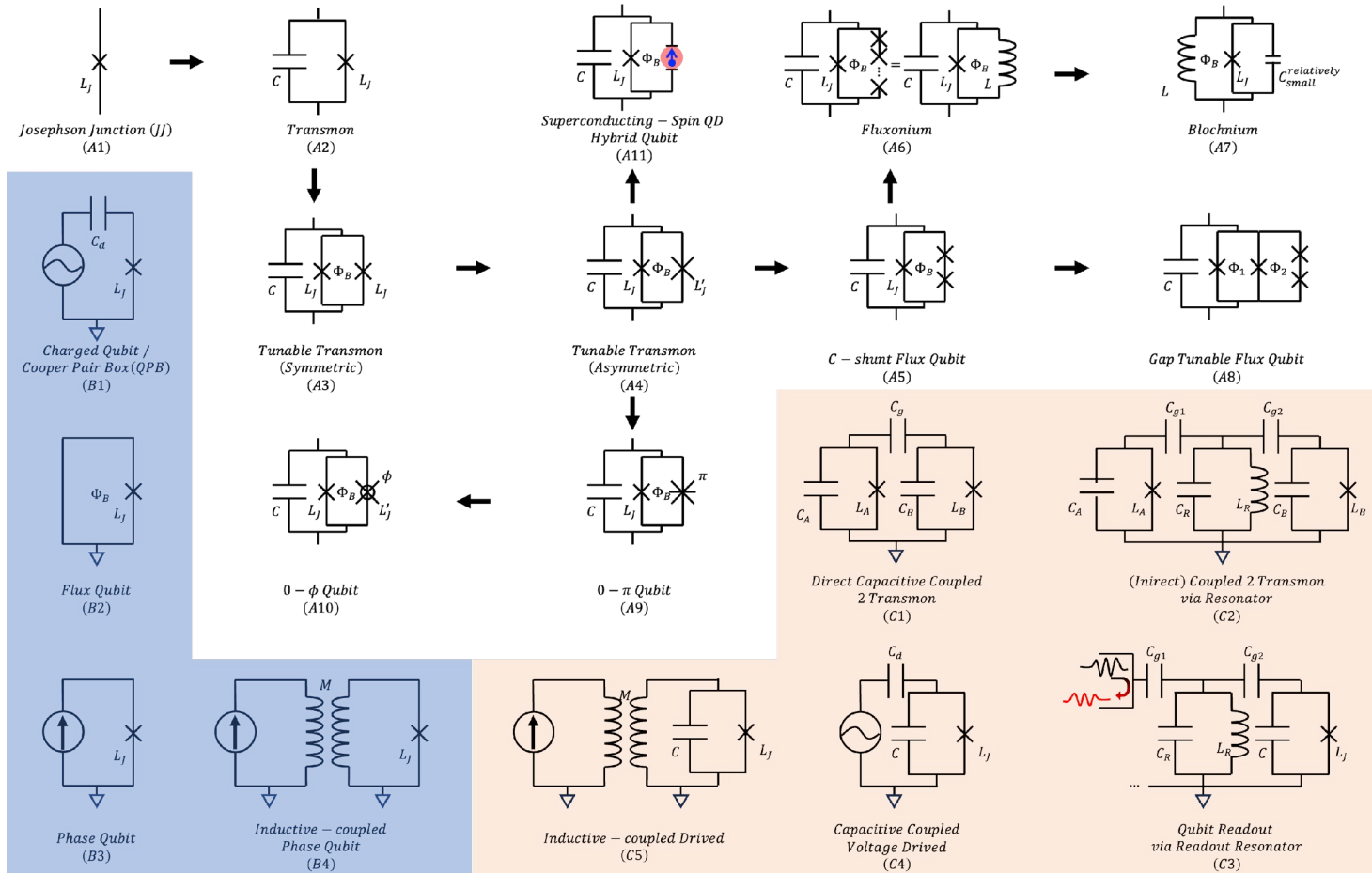


Zero Electrical Resistance



Meissner Effect

Superconducting Qubit Circuits



초전도 큐비트는 전기적 회로 형태로 구성
다른 큐비트와의 배열, 배치 문제 발생

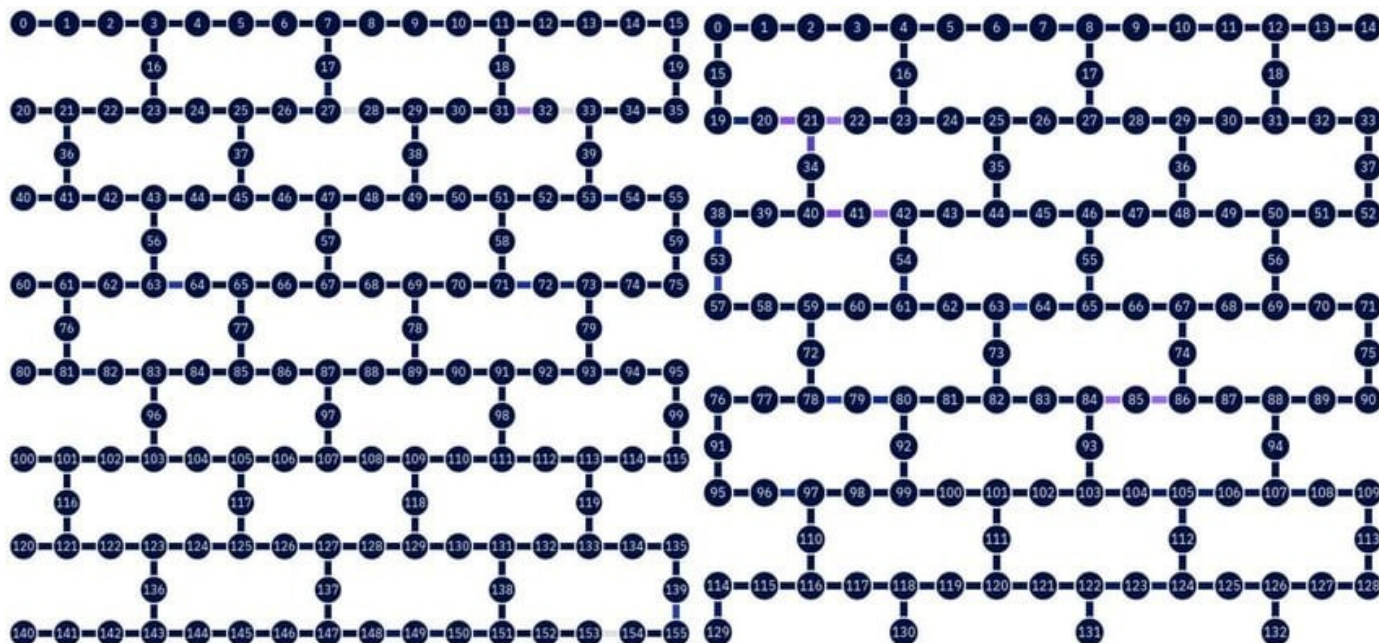
먼 큐비트와의 상호작용
다른 큐비트를 거쳐가서 형성

Properties of Superconducting Qubit

인공 원자 사용, 낮은 온도 (15mK), 열 취약성
짧은 결맞음 시간 (coherence time)

낮은 connectivity
(IBM Heron 기준-3개 큐비트)

Fast gate operating time(~ns)
Scalable한 확장성(SOTA 2025: 156Q)



3. Ion Trap Qubits

Q u a n t u m
Innovation Laboratory

Ion Traps

자연 원자(Yb, Ba) 사용, 높은 온도 (4K)

→ 높은 정확도:

긴 결맞음 시간(Coherence time)

완전 연결성(All-to-All Connectivity):

모든 큐비트끼리 직접 연결 가능

높은 진공 (10^{-11} Torr):

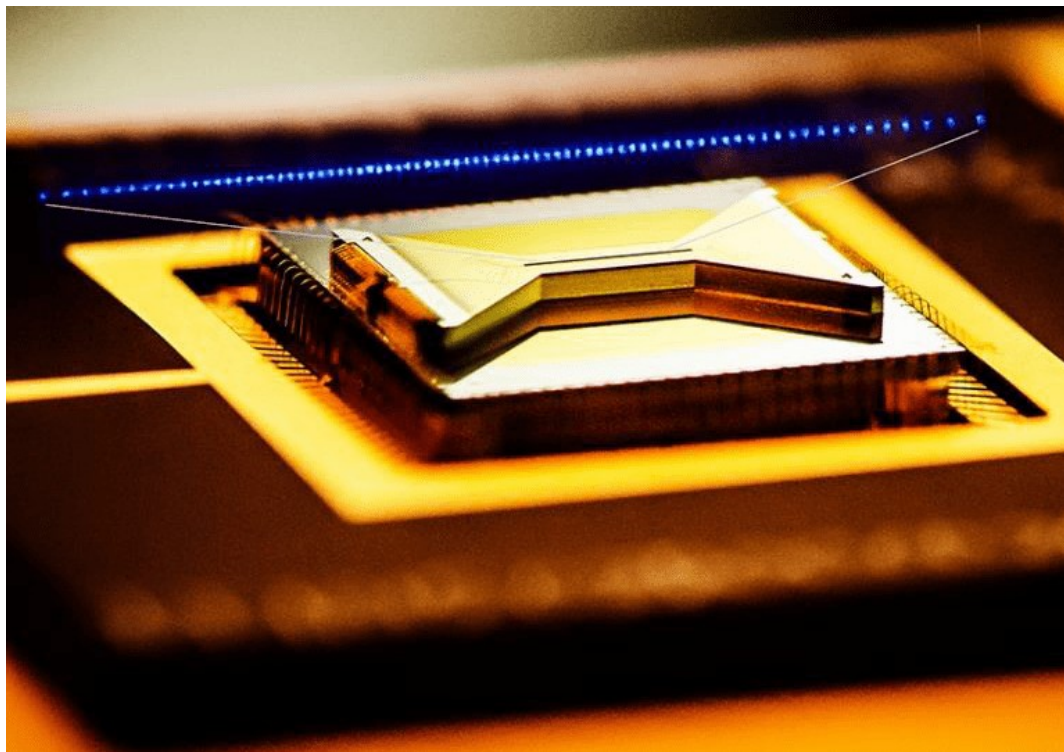
우주보다 100배 강한 진공

느린 연산 (μs):

긴 게이트 작동 시간

Non-scalable한 확장성:

한 trap에 가둘 수 있는 원자의 개수를 늘리는
난이도가 아주 높음



Ion Traps

자연 원자(Yb, Ba) 사용, 높은 온도 (4K)

→ 높은 정확도:

긴 결맞음 시간(Coherence time)

완전 연결성(All-to-All Connectivity):

모든 큐비트끼리 직접 연결 가능

높은 진공 (10^{-11} Torr):

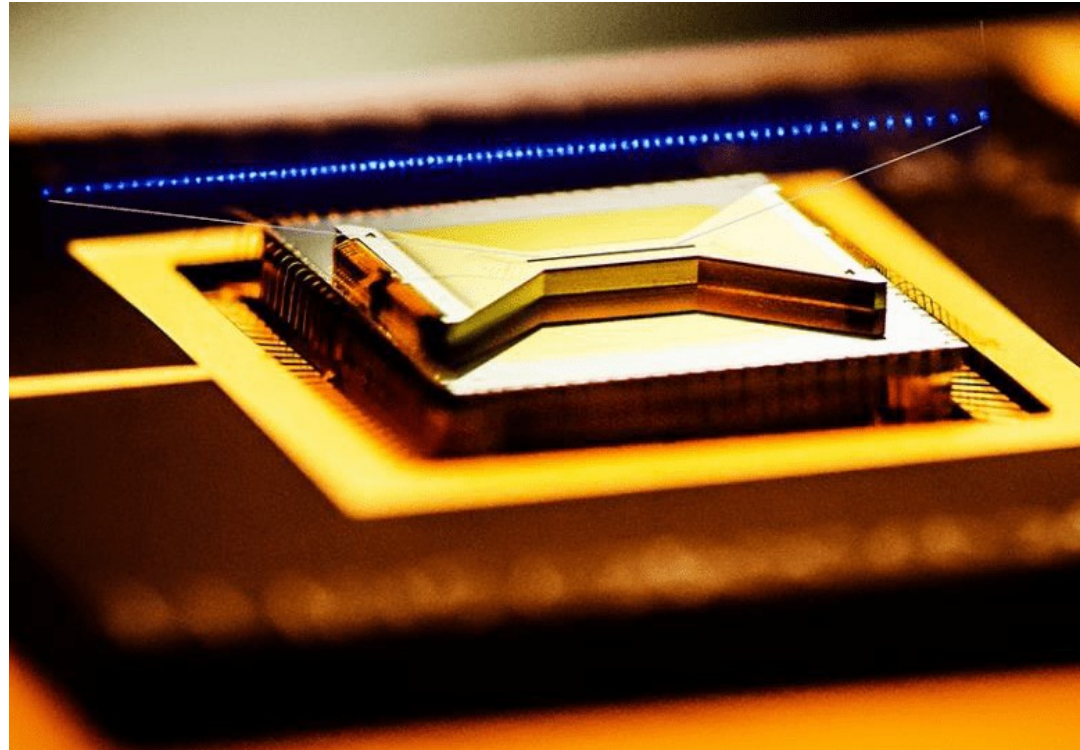
우주보다 100배 강한 진공

느린 연산 (μ s):

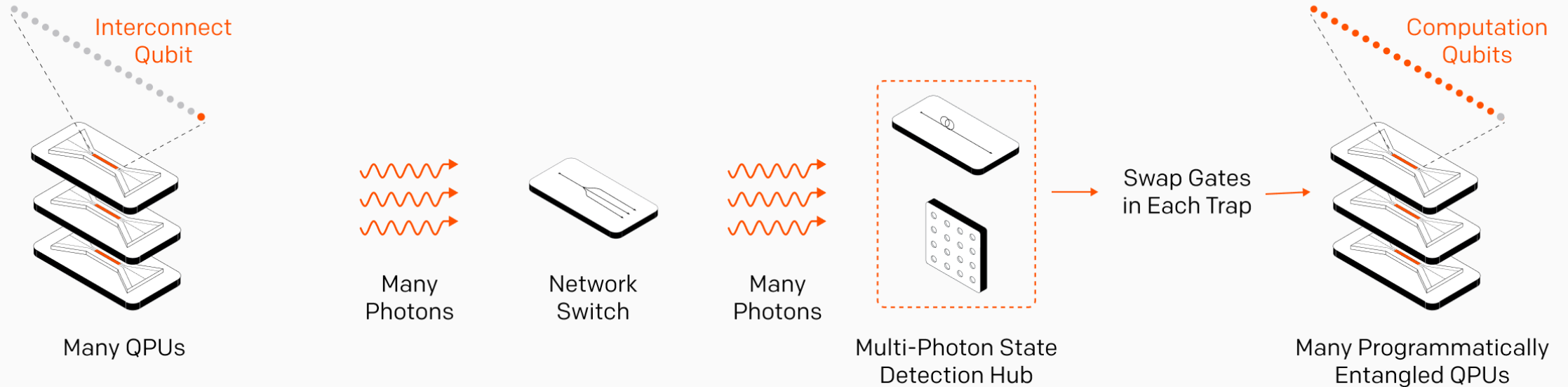
긴 게이트 작동 시간

Non-scalable한 확장성:

한 trap에 가둘 수 있는 원자의 개수를 늘리는
난이도가 아주 높음



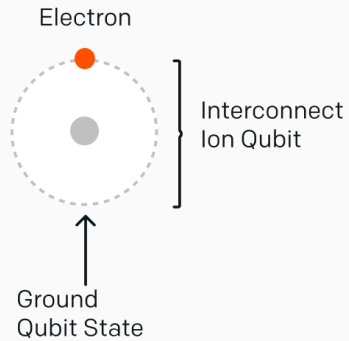
Photonic Interconnect



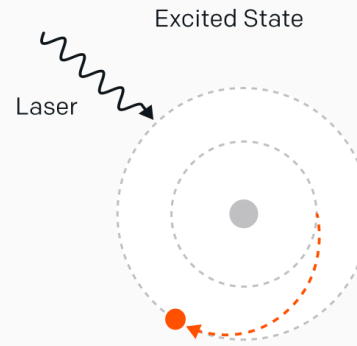
Collecting interconnect photons from many interconnect qubits across many traps, to selectively entangle quantum information across the network qubits based on the parameters of the circuit.

Photon mediates Ion trap Qubits

**Add Interconnect
Ion Qubit to Each Trap
in Network**

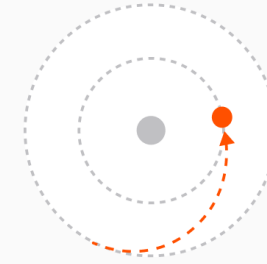


**Excite the Interconnect
Ion Qubit Electron With
a Laser**



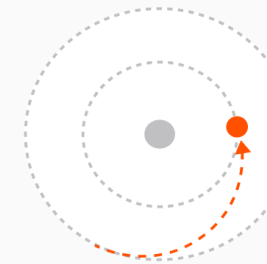
**Excited State Decays
into Superposition of
Two States**

1 Qubit State: Spin Up



+

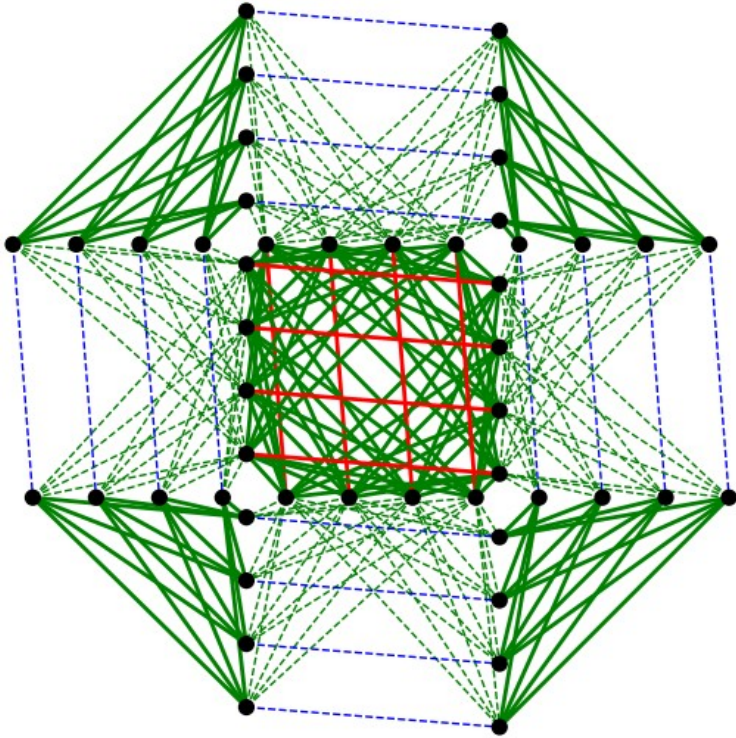
2 Qubit State: Spin Down



4. Quantum Annealer

Q u a n t u m
Innovation Laboratory

Properties of Quantum Annealer



D-Wave Zephyr topology

- 낮은 connectivity (20개 큐비트)
- Advantage2: 4400Q
- D-Wave Annealer는 Ising 문제로 변환가능한 문제만 해결가능
f QUBO

QUBO

Quadratic Unconstrained Binary Optimization

이차 무제약 이진 최적화

QUBO → Ising → Solutions

Q

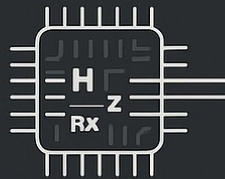
0	1	1	0
1	0	1	0
0	1	0	0
0	0	1	0



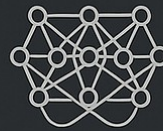
0	1	0	1

Ising mapping
 $s_i = 2x_i - 1$

QAOA



Quantum
Annealing



Simulated
Annealing



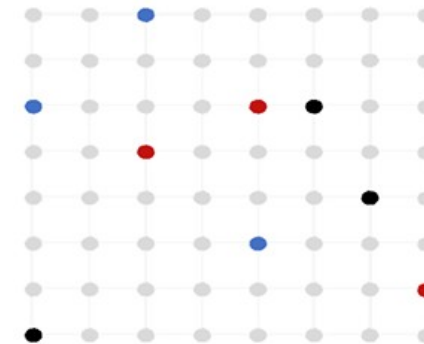
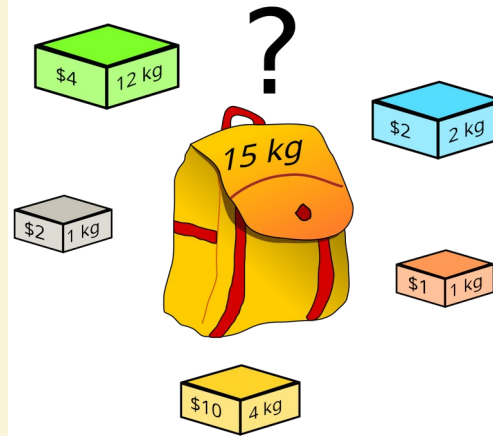
@AdnanMasood

$$H = \sum J_{ij} \sigma_i \sigma_j + h \sum \sigma_j$$

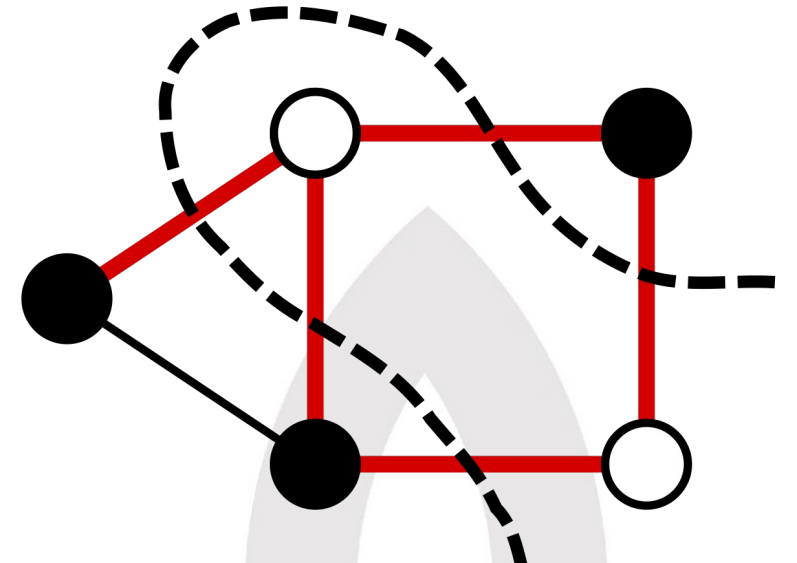
Ising Hamiltonian

Combinational Optimization Problems

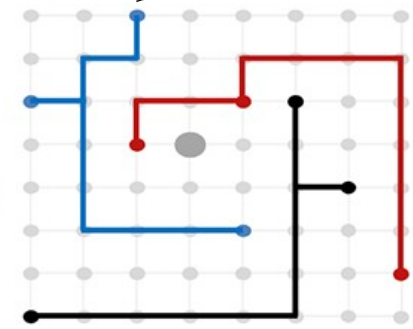
Combinational Optimization Problem 조합 최적화 문제



RSTPP 인스턴스(instance)

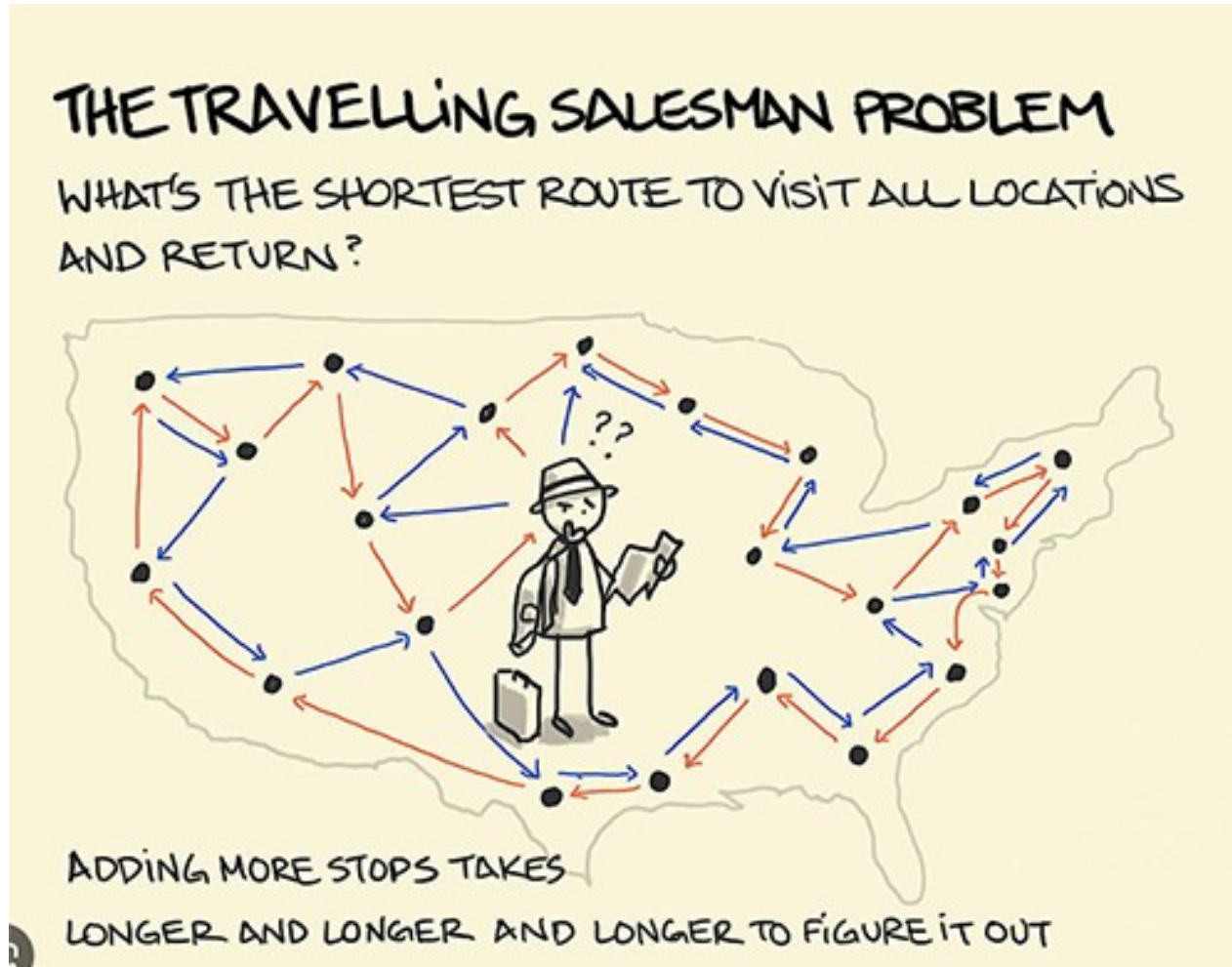


조합 최적화 알고리즘



RSTPP 해(Solution)

Travelling Salesman Problem



- **TSP**

주어진 지점(노드)를 모두 방문하는 데 최소한의 거리(시간, 비용)이 드는 방법은 무엇인가

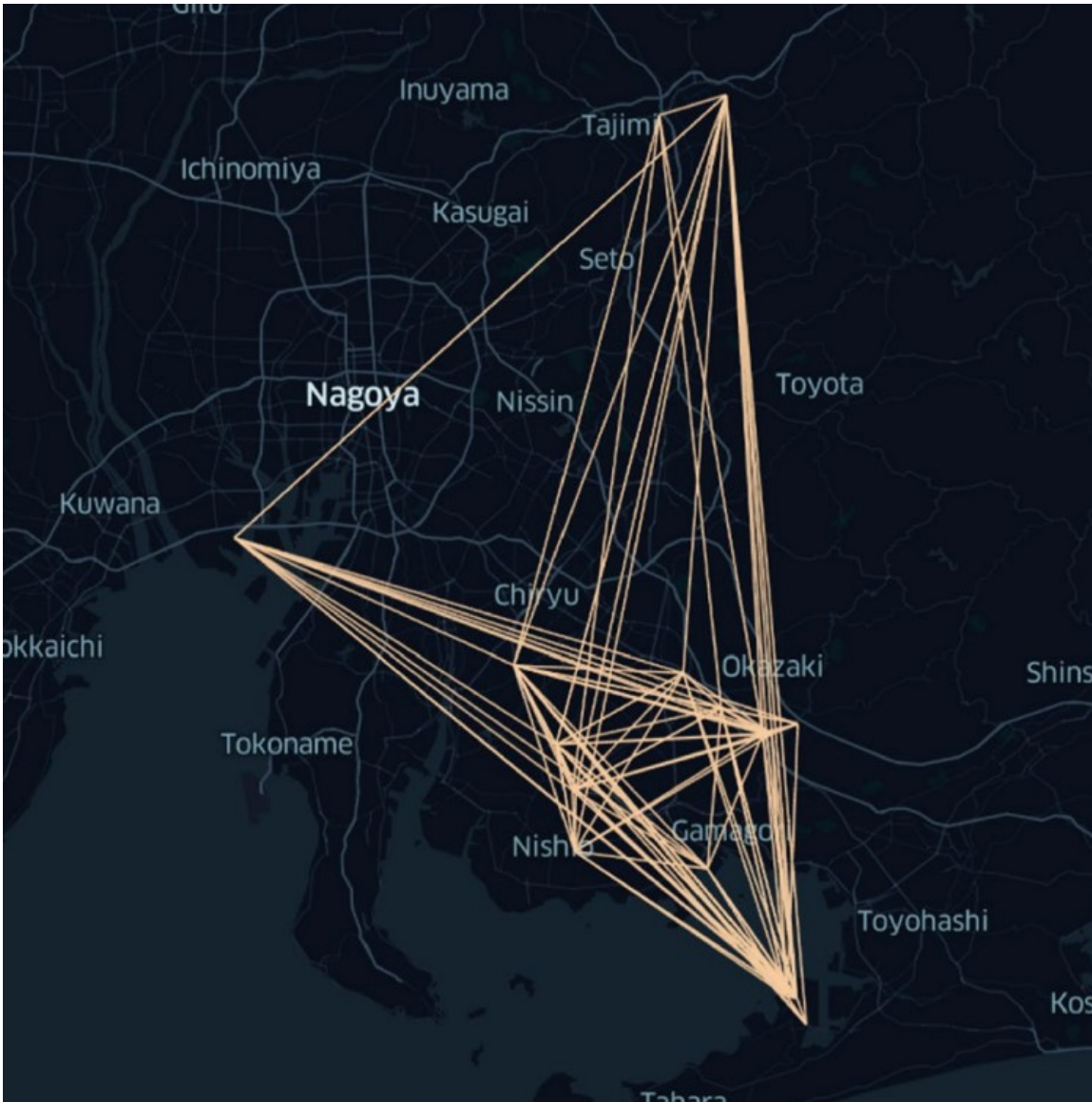
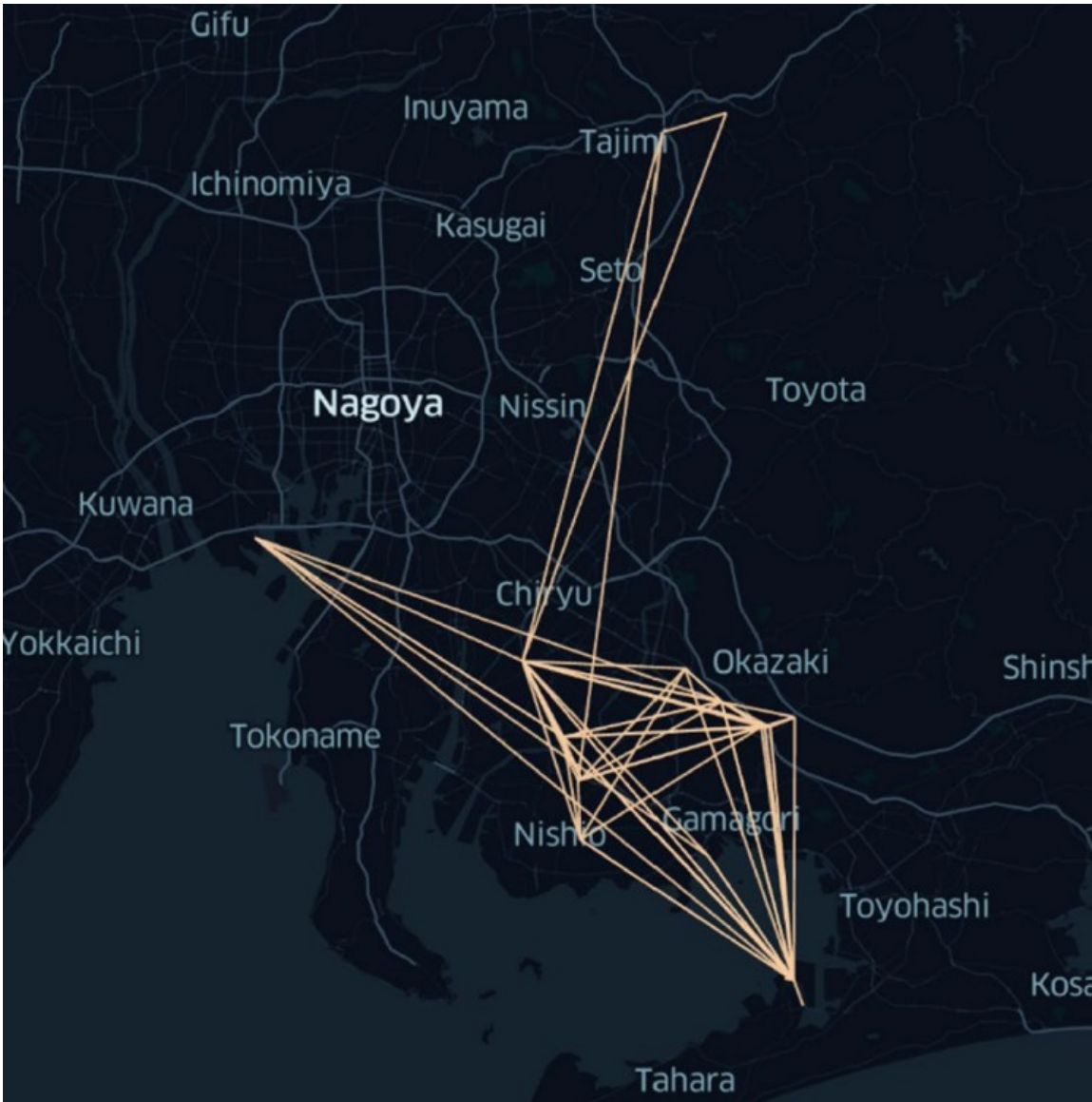
Ex)

여행 동선 최적화

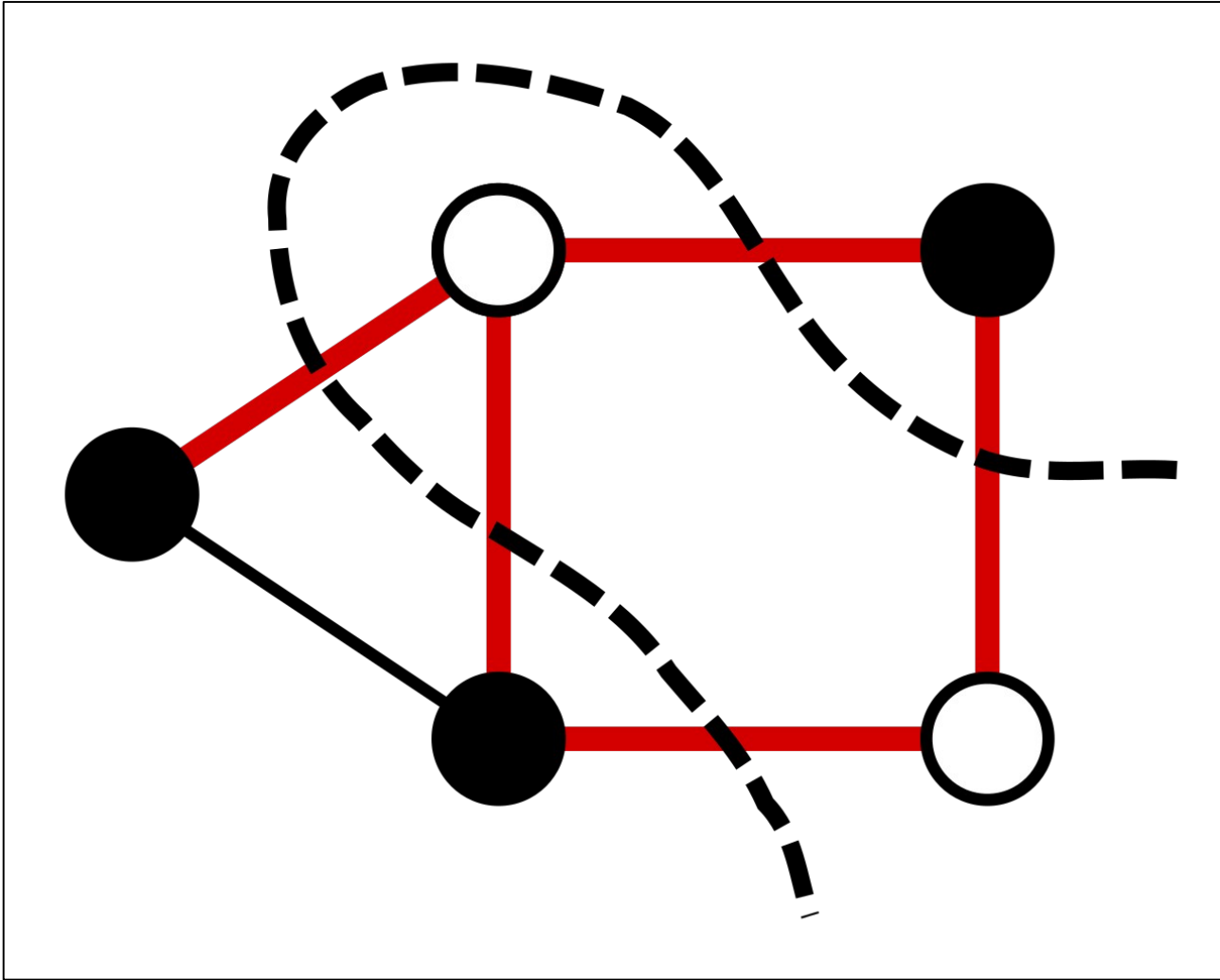
물류 최적화

...

Logistic Optimization



Max-Cut Problem



- **Max-Cut**

주어진 그래프를 두 영역으로 나눌 때, 그래프의 간선을 최대한 많이 절단하는 방법은 무엇인가

Ex)

팀 빌딩

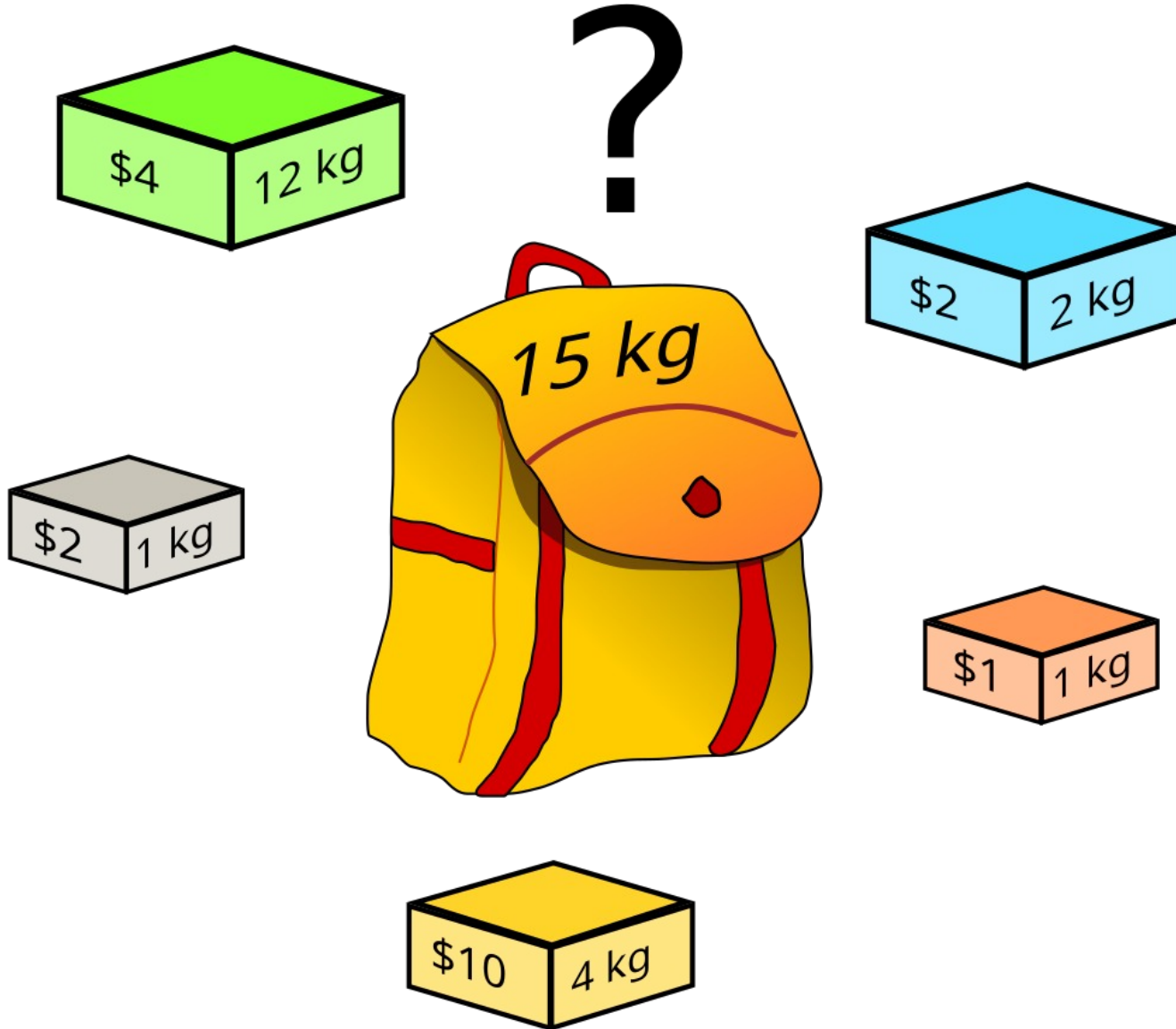
자산 포트폴리오

리스크 관리

네트워크 차단

...

Knapsack Problem

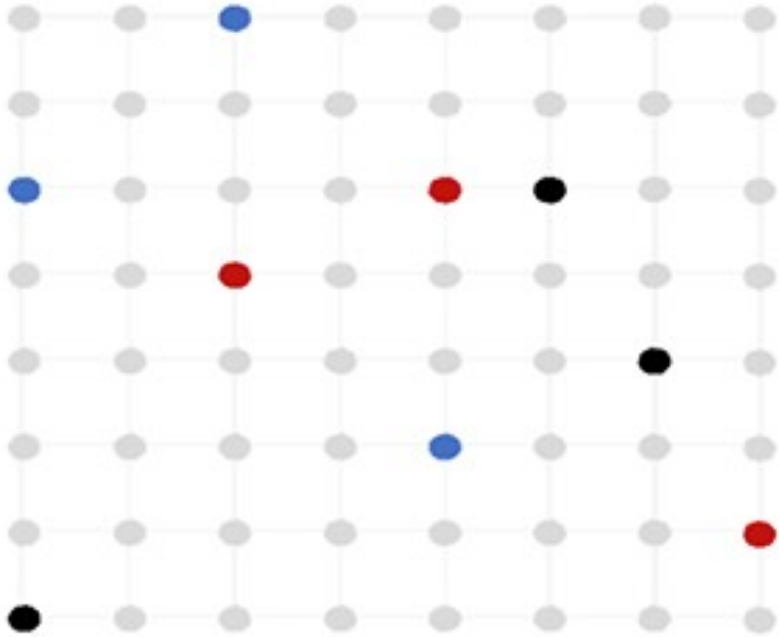


- **Knapsack Problem**

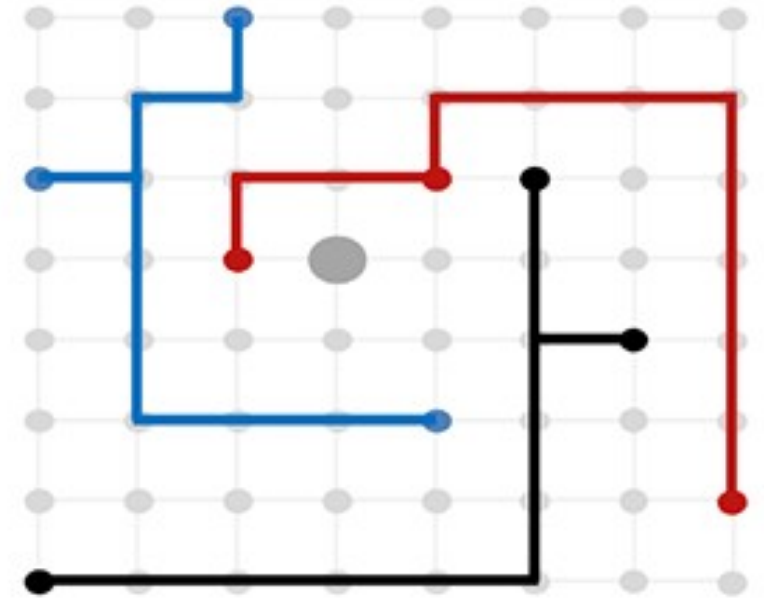
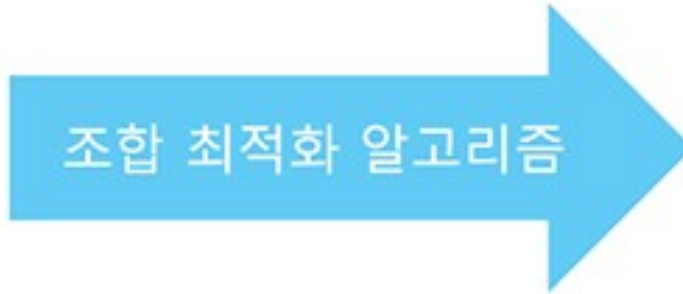
배낭의 제한조건(크기, 질량)에 가장 가치가 높은 상품을 싣는 방법은 무엇인가



Rectilinear Steiner Tree Packing Problem



RSTPP 인스턴스(instance)



RSTPP 해(Solution)

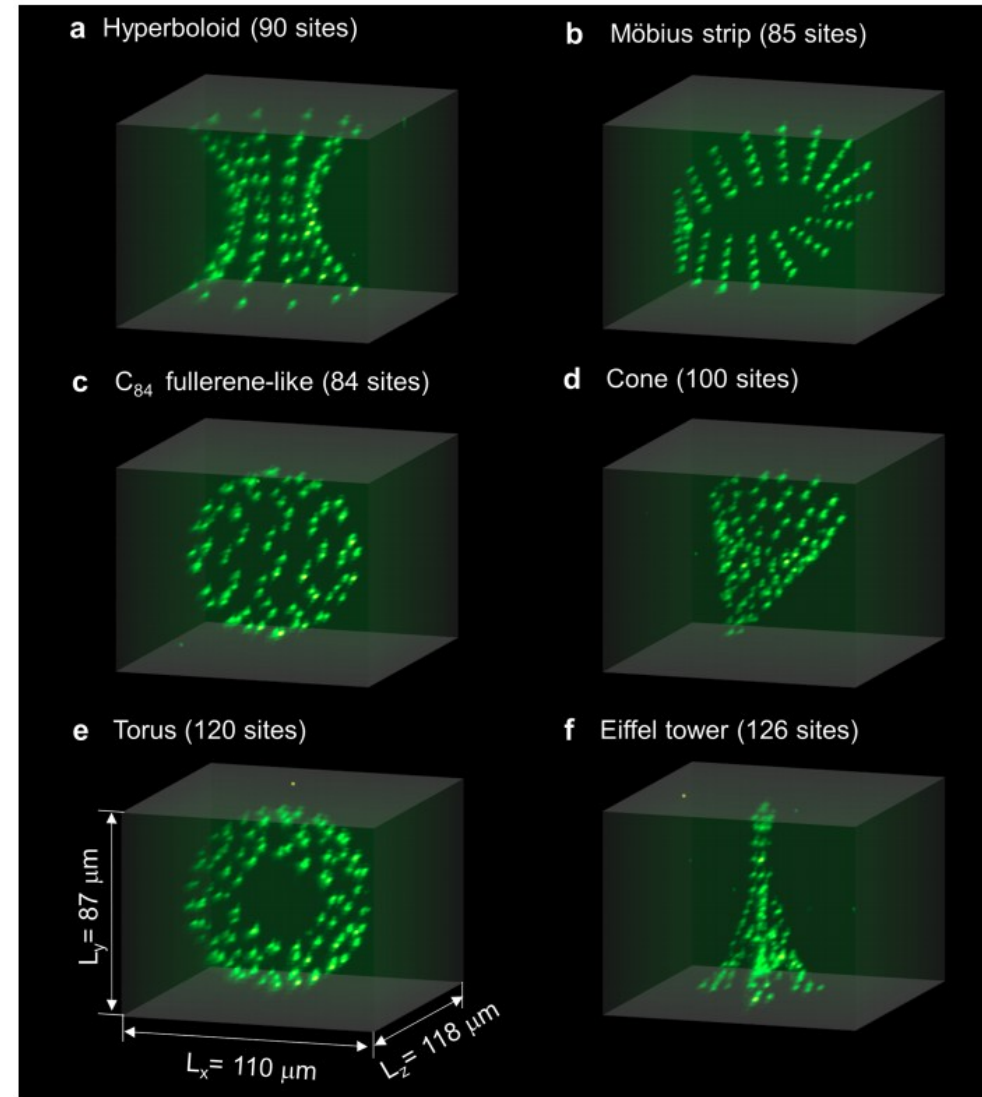
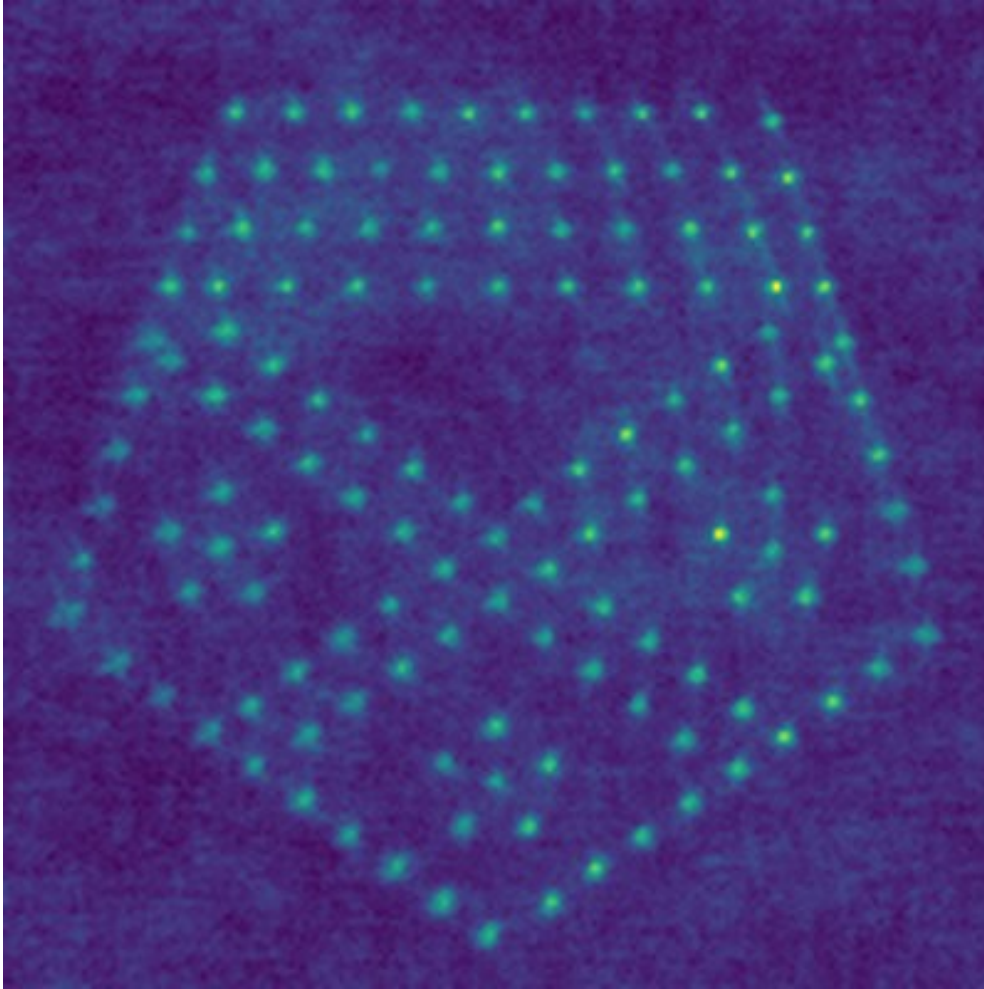
- **RSTPP**

주어진 점들을 서로 교차하는 선이 없이 최단거리로 잇는 방법은 무엇인가.

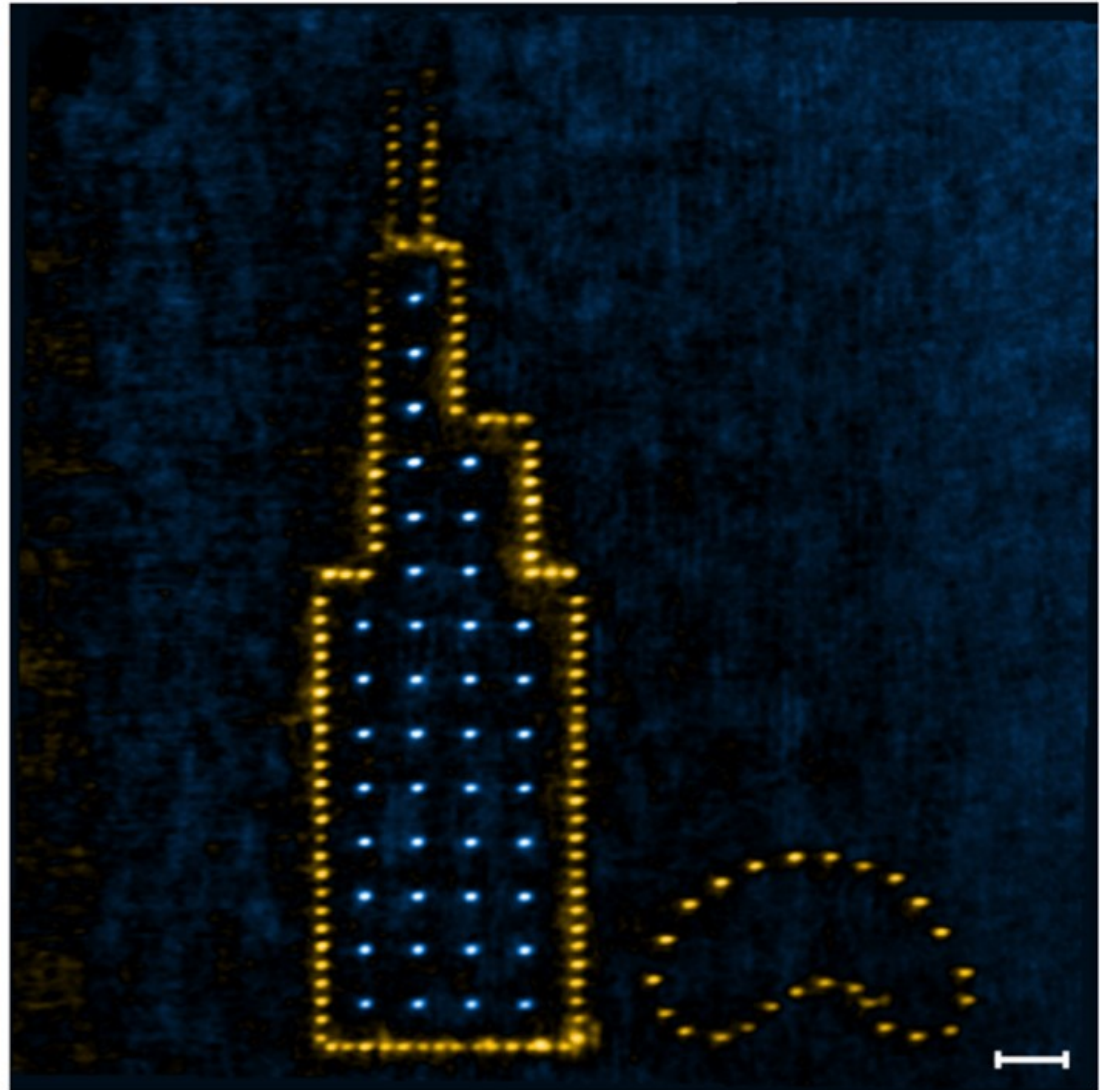
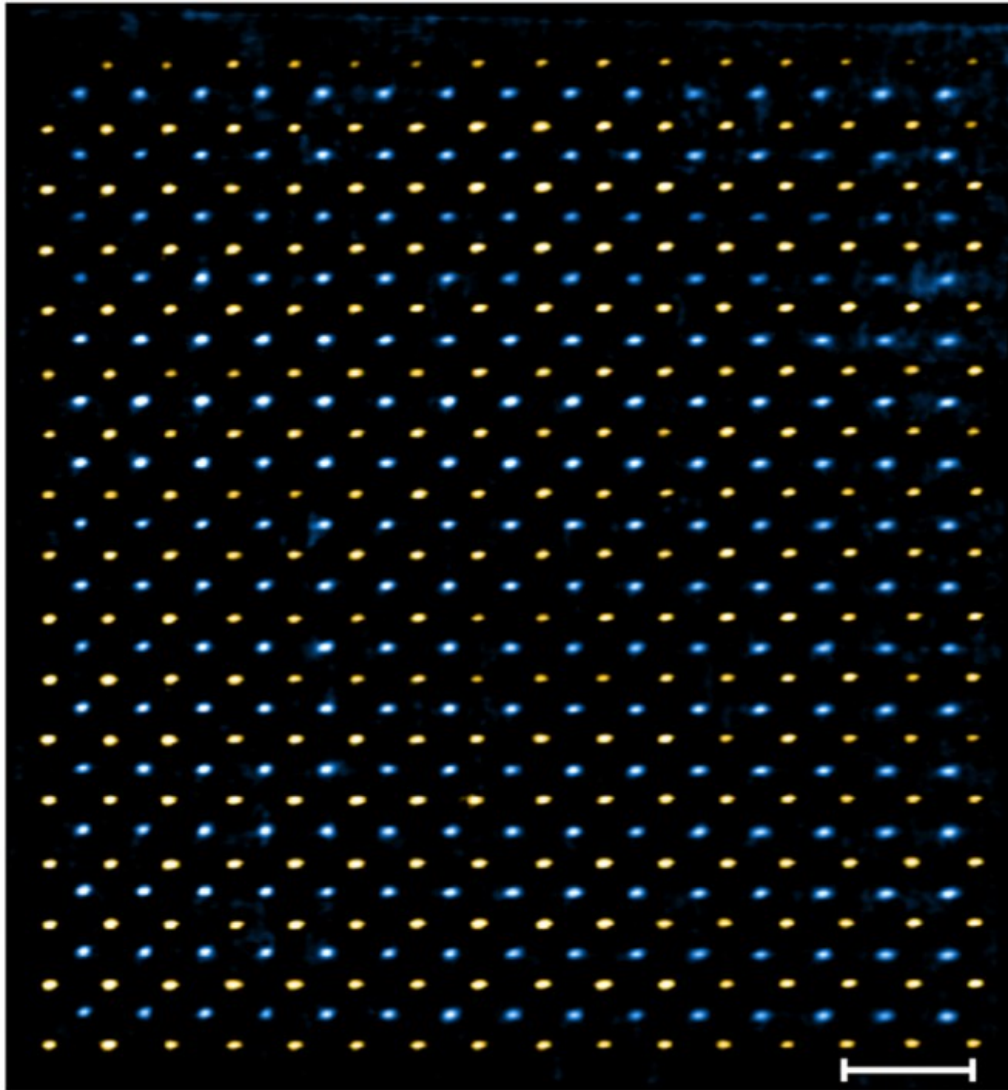
5. Neutral Atom Qubits

Q u a n t u m
Innovation Laboratory

Optical Tweezer



Optical Tweezer

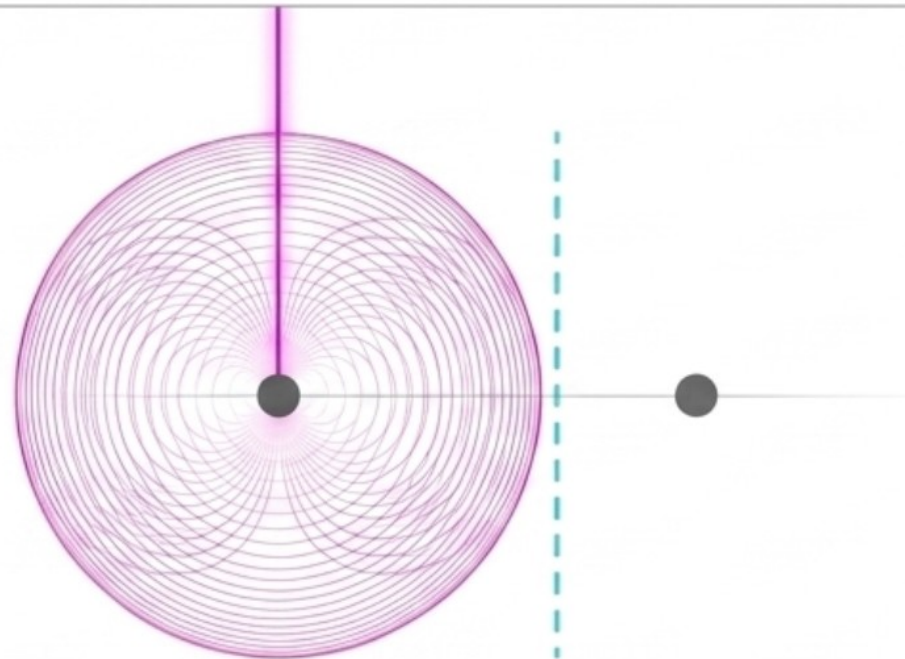


Rydberg Blockade



Step 1

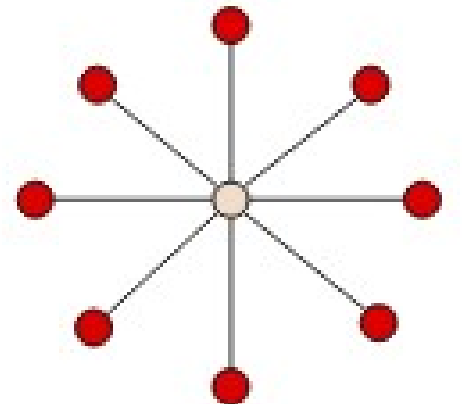
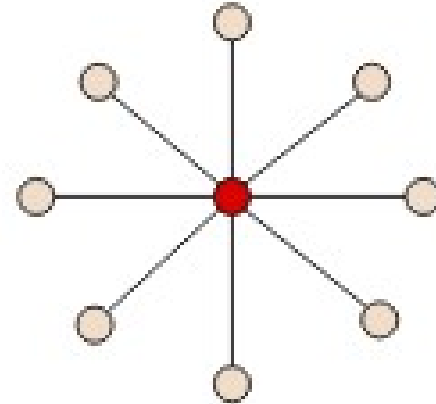
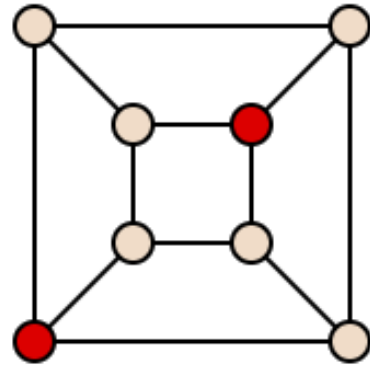
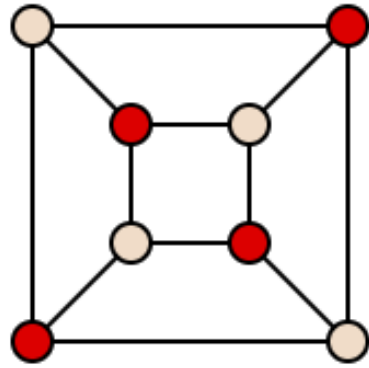
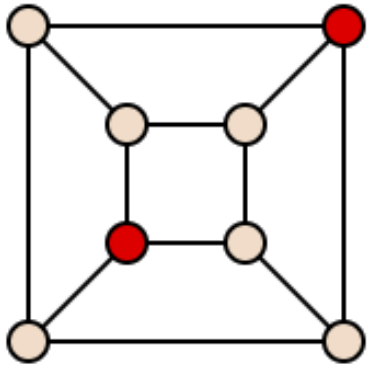
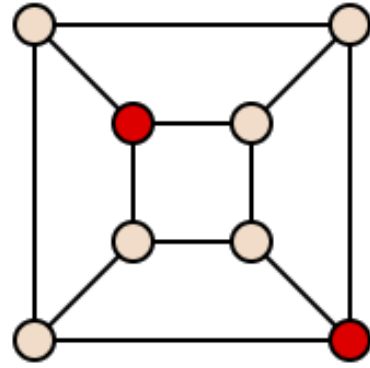
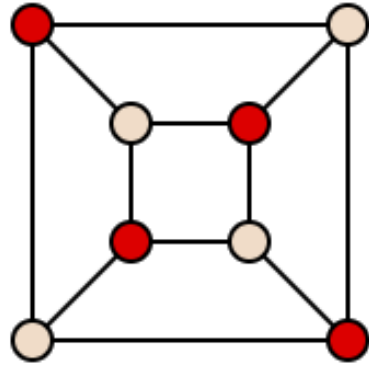
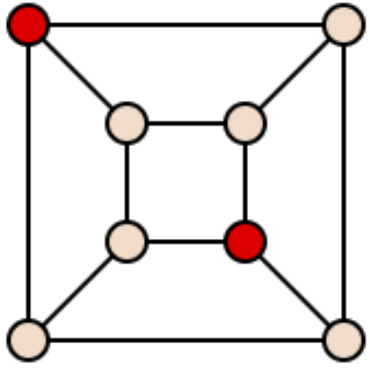
극도로 들뜬 상태 생성: 원자에 강력한 레이저를 조사하여 전자를 원자핵에서 멀리 떨어뜨린 ‘리드버그 상태’로 만듭니다. 원자 크기가 수천 배 부풀어 오릅니다.



Step 2

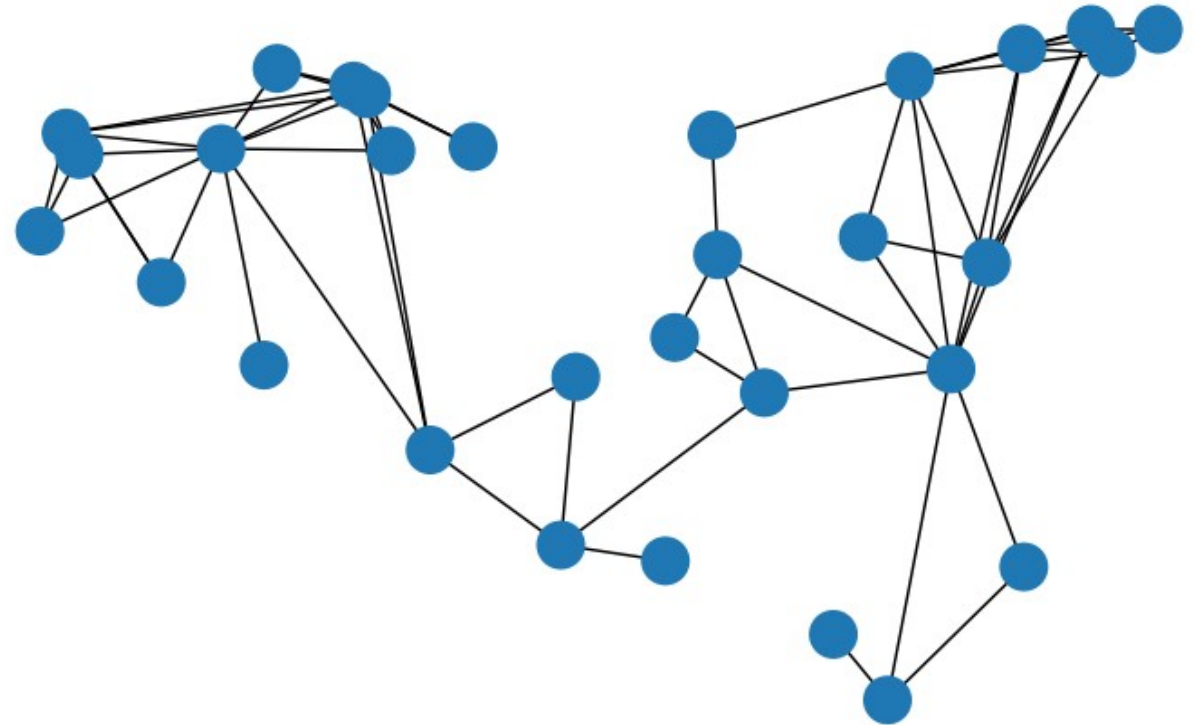
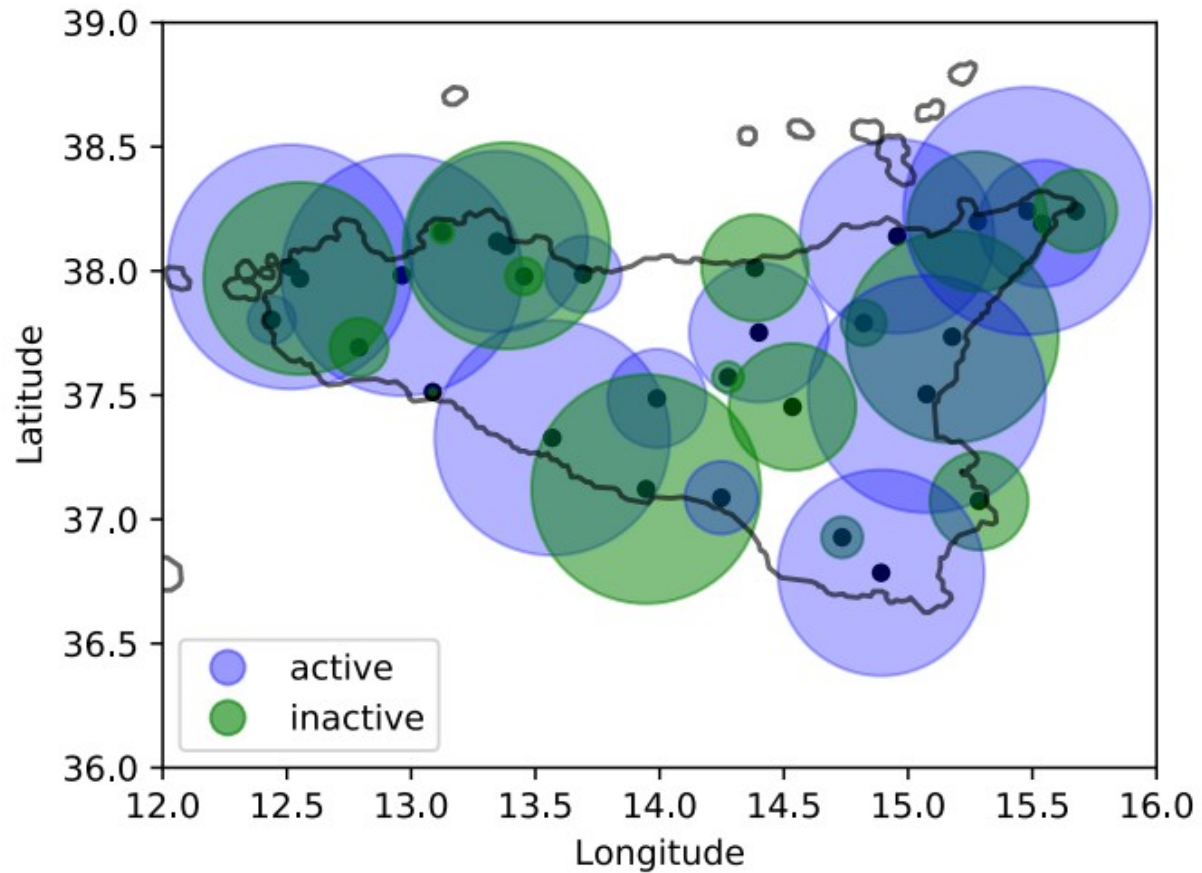
전자기장 차단 현상: 팽창한 원자는 강력한 전자기장을 띠며, 반경 수 마이크로미터 내의 다른 원자들이 리드버그 상태가 되는 것을 막습니다(Blockade). 이 ‘밀어내는 힘’을 이용해 멀리 떨어진 두 원자 간의 CNOT 게이트(양자 얽힘)를 빠르고 정확하게 수행합니다.

MIS problem

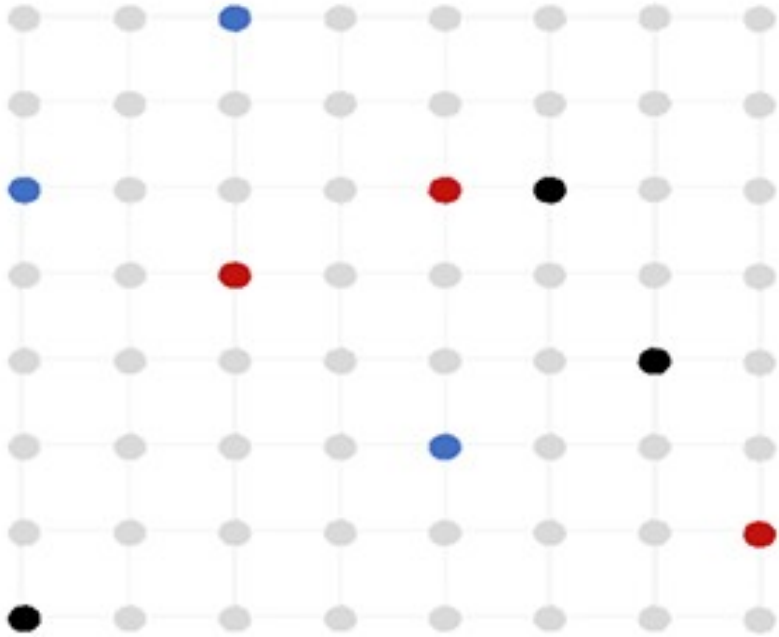


Maximum Independent Set
최대 독립 집합

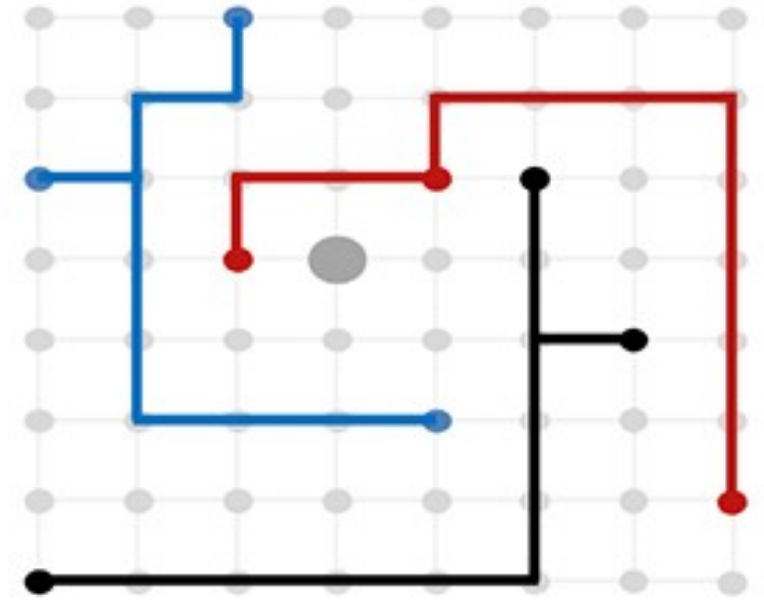
Facility Location Problem



Rectilinear Steiner Tree Packing Problem



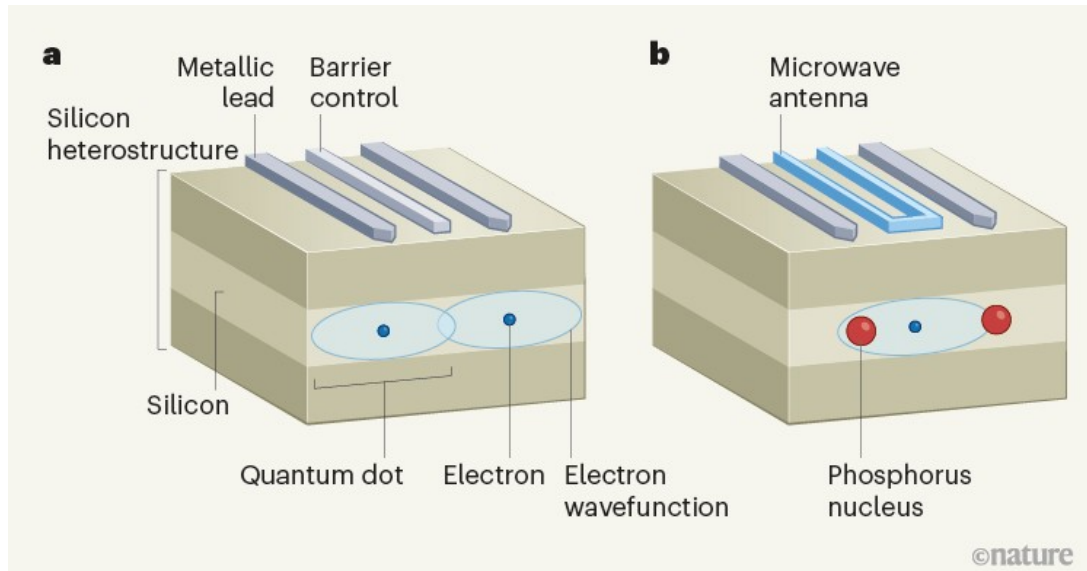
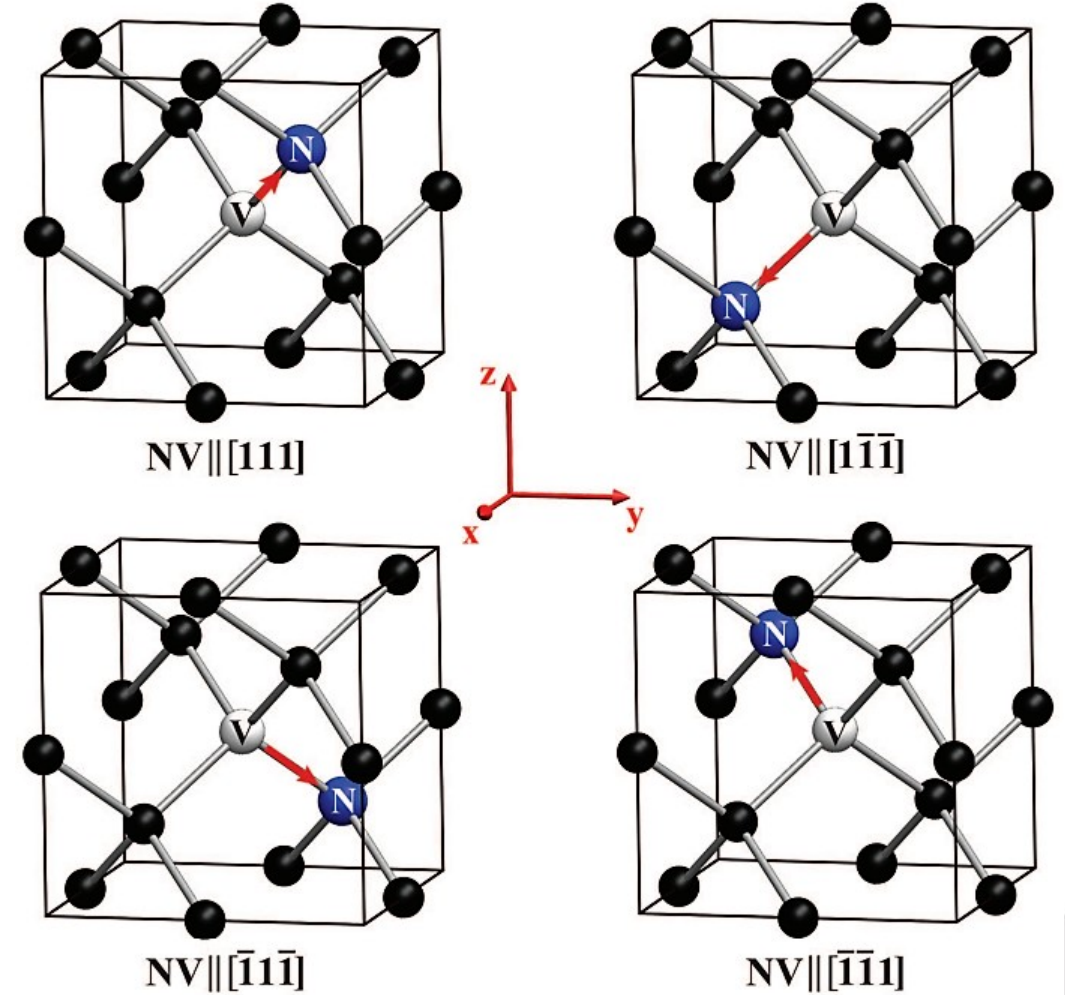
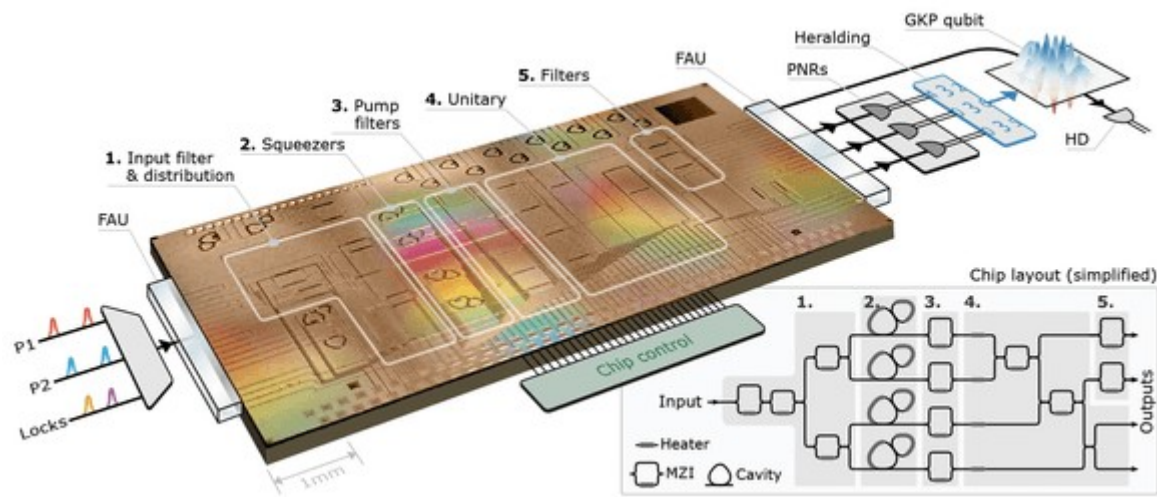
RSTPP 인스턴스(instance)



RSTPP 해(Solution)

- 주어진 문제를 푸는 방법은 유일하지 않다.
- 주어진 문제를 보유한 도구에 맞게 mapping하는 것이 양자 알고리즘 설계의 핵심

Other Qubits



Other Qubits

